**PRAKTIKUM BASIS DATA**

**MODUL 03**

**TUGAS PENDAHULUAN**


Disusun oleh:

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103122400059

**PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK**

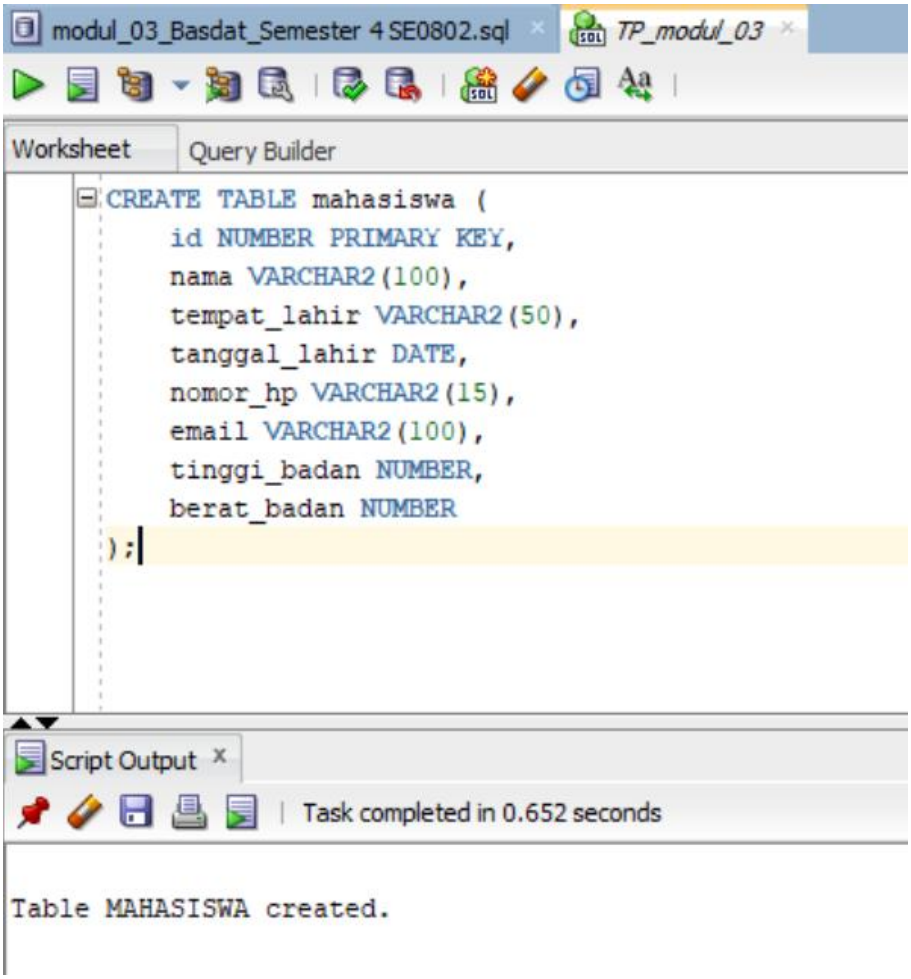
**DIREKTORAT KAMPUS PURWOKERTO**

**UNIVERSITAS TELKOM**


**2026**

**TP 1.**

1. Buat Table Mahasiswa yang beratribut (id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.)



2. **Lakukan insert data pada table nomor 1 sebanyak 15 rows.**

modul_03_Basdat_Semester 4 SE0802.sql | TP_modul_03

TP_modul_03

Worksheet | Query Builder

```
mahasiswa VALUES (3, 'Candra Pratama', 'Surabaya', TO_DATE('2002-11-22', 'YYYY-MM-DD'), '081344556677', 'candra.p@email.com', 178, 70);
mahasiswa VALUES (4, 'Dina Amelia', 'Yogyakarta', TO_DATE('2004-01-05', 'YYYY-MM-DD'), '081455667788', 'dina.a@email.com', 158, 48);
mahasiswa VALUES (5, 'Eko Saputra', 'Semarang', TO_DATE('2001-08-17', 'YYYY-MM-DD'), '081566778899', 'eko.s@email.com', 165, 60);
mahasiswa VALUES (6, 'Fani Rahmawati', 'Malang', TO_DATE('2003-12-30', 'YYYY-MM-DD'), '081677889900', 'fani.r@email.com', 162, 55);
mahasiswa VALUES (7, 'Gilang Ramadhan', 'Bogor', TO_DATE('2002-04-11', 'YYYY-MM-DD'), '081788990011', 'gilang.r@email.com', 170, 68);
mahasiswa VALUES (8, 'Hana Salsabila', 'Medan', TO_DATE('2004-07-08', 'YYYY-MM-DD'), '081899001122', 'hana.s@email.com', 155, 45);
mahasiswa VALUES (9, 'Irfan Hakim', 'Makassar', TO_DATE('2001-10-25', 'YYYY-MM-DD'), '081900112233', 'irfan.h@email.com', 175, 72);
mahasiswa VALUES (10, 'Jihan Fahira', 'Padang', TO_DATE('2003-03-14', 'YYYY-MM-DD'), '082011223344', 'jihan.f@email.com', 164, 54);
mahasiswa VALUES (11, 'Kevin Julio', 'Denpasar', TO_DATE('2002-09-09', 'YYYY-MM-DD'), '082122334455', 'kevin.j@email.com', 180, 75);
mahasiswa VALUES (12, 'Laila Fitri', 'Palembang', TO_DATE('2004-06-18', 'YYYY-MM-DD'), '082233445566', 'laila.f@email.com', 159, 49);
mahasiswa VALUES (13, 'Rizky Febian', 'Balikpapan', TO_DATE('2001-12-01', 'YYYY-MM-DD'), '082344556677', 'rizky.f@email.com', 168, 62);
mahasiswa VALUES (14, 'Nadia Vega', 'Pontianak', TO_DATE('2003-08-21', 'YYYY-MM-DD'), '082455667788', 'nadia.v@email.com', 161, 52);
mahasiswa VALUES (15, 'Tegar Septian', 'Banjarmasin', TO_DATE('2002-02-28', 'YYYY-MM-DD'), '082566778899', 'tegar.s@email.com', 174, 69);
```

Script Output | Task completed in 0.044 seconds

```
Table MAHASISWA created.

1 row inserted.
```

modul_03_Basdat_Semester 4 SE0802.sql | TP_modul_03

Worksheet | Query Builder

```
INSERT INTO mahasiswa VALUES (5, 'Eko Saputra', 'Semarang', TO_DATE('2001-08-17', 'YYYY-MM-DD'), '081566778899', 'ek
INSERT INTO mahasiswa VALUES (6, 'Fani Rahmawati', 'Malang', TO_DATE('2003-12-30', 'YYYY-MM-DD'), '081677889900', 'f
INSERT INTO mahasiswa VALUES (7, 'Gilang Ramadhan', 'Bogor', TO_DATE('2002-04-11', 'YYYY-MM-DD'), '081788990011', 'g
INSERT INTO mahasiswa VALUES (8, 'Hana Salsabila', 'Medan', TO_DATE('2004-07-08', 'YYYY-MM-DD'), '081899001122', 'ha
INSERT INTO mahasiswa VALUES (9, 'Irfan Hakim', 'Makassar', TO_DATE('2001-10-25', 'YYYY-MM-DD'), '081900112233', 'ir
INSERT INTO mahasiswa VALUES (10, 'Jihan Fahira', 'Padang', TO_DATE('2003-03-14', 'YYYY-MM-DD'), '082011223344', 'ji
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.005 seconds

| | ID | NAMA | TEMPAT_LAHIR | TANGGAL_LAHIR | NOMOR_HP | EMAIL | TINGGI_BADAN | BERAT_BADAN |
|---|---|---|---|---|---|---|---|---|
| 1 | 15 | Tegar Septian | Banjarmasin | 28-FEB-02 | 082566778899 | tegar.s@email.com | 174 | 69 |
| 2 | 1 | Andra Wijaya | Bandung | 15-FEB-04 | 081122334455 | andra.w@email.com | 172 | 65 |
| 3 | 2 | Bunga Lestari | Jakarta | 10-MAY-03 | 081233445566 | bunga.l@email.com | 160 | 50 |
| 4 | 3 | Candra Pratama | Surabaya | 22-NOV-02 | 081344556677 | candra.p@email.com | 178 | 70 |
| 5 | 4 | Dina Amelia | Yogyakarta | 05-JAN-04 | 081455667788 | dina.a@email.com | 158 | 48 |
| 6 | 5 | Eko Saputra | Semarang | 17-AUG-01 | 081566778899 | eko.s@email.com | 165 | 60 |
| 7 | 6 | Fani Rahmawati | Malang | 30-DEC-03 | 081677889900 | fani.r@email.com | 162 | 55 |
| 8 | 7 | Gilang Ramadhan | Bogor | 11-APR-02 | 081788990011 | gilang.r@email.com | 170 | 68 |
| 9 | 8 | Hana Salsabila | Medan | 08-JUL-04 | 081899001122 | hana.s@email.com | 155 | 45 |
| 10 | 9 | Irfan Hakim | Makassar | 25-OCT-01 | 081900112233 | irfan.h@email.com | 175 | 72 |
| 11 | 10 | Jihan Fahira | Padang | 14-MAR-03 | 082011223344 | jihan.f@email.com | 164 | 54 |
| 12 | 11 | Kevin Julio | Denpasar | 09-SEP-02 | 082122334455 | kevin.j@email.com | 180 | 75 |
| 13 | 12 | Laila Fitri | Palembang | 18-JUN-04 | 082233445566 | laila.f@email.com | 159 | 49 |
| 14 | 13 | Rizky Febian | Balikpapan | 01-DEC-01 | 082344556677 | rizky.f@email.com | 168 | 62 |
| 15 | 14 | Nadia Vega | Pontianak | 21-AUG-03 | 082455667788 | nadia.v@email.com | 161 | 52 |

3. **Lakukan Fungsi**

   a. **Fungsi Karakter dan String :**

```sql
SELECT
    nama,
    LOWER(nama) AS huruf_kecil,
    UPPER(nama) AS huruf_besar,
    INITCAP(nama) AS kapital_awal,
    CONCAT(nama, ' - Mahasiswa Aktif') AS gabungan,
    LENGTH(nama) AS panjang_nama,
    SUBSTR(nama, 1, 5) AS potong_nama,
    LPAD(nama, 25, '#') AS lpad_nama,
    RPAD(nama, 25, '#') AS rpad_nama,
    LTRIM('    ' || nama) AS ltrim_nama,
    RTRIM(nama || '    ') AS rtrim_nama,
    INSTR(nama, 'i') AS posisi_i,
    REPLACE(nama, 'a', '4') AS ganti_a
FROM mahasiswa;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.014 seconds

| | NAMA | HURUF_KECIL | HURUF_BESAR | KAPITAL_AWAL | GABUNGAN | PANJANG_NAMA | POTONG_NAMA |
|---|---|---|---|---|---|---|---|
| 1 | Tegar Septian | tegar septian | TEGAR SEPTIAN | Tegar Septian | Tegar Septian - Mahasiswa Aktif | 13 | Tegar |
| 2 | Andra Wijaya | andra wijaya | ANDRA WIJAYA | Andra Wijaya | Andra Wijaya - Mahasiswa Aktif | 12 | Andra |
| 3 | Bunga Lestari | bunga lestari | BUNGA LESTARI | Bunga Lestari | Bunga Lestari - Mahasiswa Aktif | 13 | Bunga |
| 4 | Candra Pratama | candra pratama | CANDRA PRATAMA | Candra Pratama | Candra Pratama - Mahasiswa Aktif | 14 | Candr |
| 5 | Dina Amelia | dina amelia | DINA AMELIA | Dina Amelia | Dina Amelia - Mahasiswa Aktif | 11 | Dina |
| 6 | Eko Saputra | eko saputra | EKO SAPUTRA | Eko Saputra | Eko Saputra - Mahasiswa Aktif | 11 | Eko S |
| 7 | Fani Rahmawati | fani rahmawati | FANI RAHMAWATI | Fani Rahmawati | Fani Rahmawati - Mahasiswa Aktif | 14 | Fani |
| 8 | Gilang Ramadhan | gilang ramadhan | GILANG RAMADHAN | Gilang Ramadhan | Gilang Ramadhan - Mahasiswa Aktif | 15 | Gilan |
| 9 | Hana Salsabila | hana salsabila | HANA SALSABILA | Hana Salsabila | Hana Salsabila - Mahasiswa Aktif | 14 | Hana |
| 10 | Irfan Hakim | irfan hakim | IRFAN HAKIM | Irfan Hakim | Irfan Hakim - Mahasiswa Aktif | 11 | Irfan |
| 11 | Jihan Fahira | jihan fahira | JIHAN FAHIRA | Jihan Fahira | Jihan Fahira - Mahasiswa Aktif | 12 | Jihan |
| 12 | Kevin Julio | kevin julio | KEVIN JULIO | Kevin Julio | Kevin Julio - Mahasiswa Aktif | 11 | Kevin |
| 13 | Laila Fitri | laila fitri | LAILA FITRI | Laila Fitri | Laila Fitri - Mahasiswa Aktif | 11 | Laila |
| 14 | Rizky Febian | rizky febian | RIZKY FEBIAN | Rizky Febian | Rizky Febian - Mahasiswa Aktif | 12 | Rizky |
| 15 | Nadia Vega | nadia vega | NADIA VEGA | Nadia Vega | Nadia Vega - Mahasiswa Aktif | 10 | Nadia |

| LPAD_NAMA | RPAD_NAMA | LTRIM_NAMA | RTRIM_NAMA | POSISI_I | GANTI_A |
|---|---|---|---|---|---|
| ############Tegar Septian | Tegar Septian############ | Tegar Septian | Tegar Septian | 11 | Teg4r Septi4n |
| #############Andra Wijaya | Andra Wijaya############# | Andra Wijaya | Andra Wijaya | 8 | Andr4 Wij4y4 |
| ############Bunga Lestari | Bunga Lestari############ | Bunga Lestari | Bunga Lestari | 13 | Bung4 Lest4ri |
| ###########Candra Pratama | Candra Pratama########### | Candra Pratama | Candra Pratama | 0 | C4ndr4 Pr4t4m4 |
| #############Dina Amelia | Dina Amelia############# | Dina Amelia | Dina Amelia | 2 | Din4 Ameli4 |
| #############Eko Saputra | Eko Saputra############# | Eko Saputra | Eko Saputra | 0 | Eko S4putr4 |
| ###########Fani Rahmawati | Fani Rahmawati########### | Fani Rahmawati | Fani Rahmawati | 4 | F4ni R4hm4w4ti |
| ##########Gilang Ramadhan | Gilang Ramadhan########## | Gilang Ramadhan | Gilang Ramadhan | 2 | Gil4ng R4m4dh4n |
| ###########Hana Salsabila | Hana Salsabila########### | Hana Salsabila | Hana Salsabila | 12 | H4n4 S4ls4bil4 |
| ##############Irfan Hakim | Irfan Hakim############## | Irfan Hakim | Irfan Hakim | 10 | Irf4n H4kim |
| #############Jihan Fahira | Jihan Fahira############# | Jihan Fahira | Jihan Fahira | 2 | Jih4n F4hir4 |
| #############Kevin Julio | Kevin Julio############## | Kevin Julio | Kevin Julio | 4 | Kevin Julio |
| #############Laila Fitri | Laila Fitri############## | Laila Fitri | Laila Fitri | 3 | L4il4 Fitri |
| ############Rizky Febian | Rizky Febian############ | Rizky Febian | Rizky Febian | 2 | Rizky Febi4n |
| ###############Nadia Vega | Nadia Vega############### | Nadia Vega | Nadia Vega | 4 | N4di4 Veg4 |

**b. fungsi tanggal dan waktu**

```sql
SELECT
    tanggal_lahir,
    SYSDATE AS hari_ini,
    ADD_MONTHS(tanggal_lahir, 3) AS tambah_bulan,
    LAST_DAY(tanggal_lahir) AS akhir_bulan,
    NEXT_DAY(tanggal_lahir, 'MONDAY') AS senin_berikut,
    MONTHS_BETWEEN(SYSDATE, tanggal_lahir) AS selisih_bulan,
    ROUND(SYSDATE, 'YYYY') AS pembulatan_tahun
FROM mahasiswa;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 15 in 0.01 seconds

| | TANGGAL_LAHIR | HARI_INI | TAMBAH_BULAN | AKHIR_BULAN | SENIN_BERIKUT | SELISIH_BULAN | PEMBULATAN_TAH |
|---|---|---|---|---|---|---|---|
| 1 | 28-FEB-02 | 25-MAR-26 | 31-MAY-02 | 28-FEB-02 | 04-MAR-02 | 288.9224070340501792114695340501792114 7 | 01-JAN-26 |
| 2 | 15-FEB-04 | 25-MAR-26 | 15-MAY-04 | 29-FEB-04 | 16-FEB-04 | 265.3417618727598566308243727598566308 24 | 01-JAN-26 |
| 3 | 10-MAY-03 | 25-MAR-26 | 10-AUG-03 | 31-MAY-03 | 12-MAY-03 | 274.5030521953405017921146953405017921 15 | 01-JAN-26 |
| 4 | 22-NOV-02 | 25-MAR-26 | 22-FEB-03 | 30-NOV-02 | 25-NOV-02 | 280.1159554211469534050179211469534050 18 | 01-JAN-26 |
| 5 | 05-JAN-04 | 25-MAR-26 | 05-APR-04 | 31-JAN-04 | 12-JAN-04 | 266.6643425179211469534050179211469534 05 | 01-JAN-26 |
| 6 | 17-AUG-01 | 25-MAR-26 | 17-NOV-01 | 31-AUG-01 | 20-AUG-01 | 295.2772457437275985663082437275985663 08 | 01-JAN-26 |
| 7 | 30-DEC-03 | 25-MAR-26 | 30-MAR-04 | 31-DEC-03 | 05-JAN-04 | 266.8578909050179211469534050179211469 53 | 01-JAN-26 |
| 8 | 11-APR-02 | 25-MAR-26 | 11-JUL-02 | 30-APR-02 | 15-APR-02 | 287.4707941308243727598566308243727598 57 | 01-JAN-26 |
| 9 | 08-JUL-04 | 25-MAR-26 | 08-OCT-04 | 31-JUL-04 | 12-JUL-04 | 260.5675683243727598566308243727598566 31 | 01-JAN-26 |
| 10 | 25-OCT-01 | 25-MAR-26 | 25-JAN-02 | 31-OCT-01 | 29-OCT-01 | 293 | 01-JAN-26 |
| 11 | 14-MAR-03 | 25-MAR-26 | 14-JUN-03 | 31-MAR-03 | 17-MAR-03 | 276.3740199372759856630824372759856630 82 | 01-JAN-26 |
| 12 | 09-SEP-02 | 25-MAR-26 | 09-DEC-02 | 30-SEP-02 | 16-SEP-02 | 282.5353102598566308243727598566308243 73 | 01-JAN-26 |
| 13 | 18-JUN-04 | 25-MAR-26 | 18-SEP-04 | 30-JUN-04 | 21-JUN-04 | 261.2449876792114695340501792114695340 5 | 01-JAN-26 |
| 14 | 01-DEC-01 | 25-MAR-26 | 01-MAR-02 | 31-DEC-01 | 03-DEC-01 | 291.7933747759856630824372759856630824 37 | 01-JAN-26 |
| 15 | 21-AUG-03 | 25-MAR-26 | 21-NOV-03 | 31-AUG-03 | 25-AUG-03 | 271.1482134856630824372759856630824372 76 | 01-JAN-26 |

**c. Fungsi Konversi**

```sql
SELECT
    nama,
    TO_CHAR(tanggal_lahir, 'DD-MM-YYYY') AS tanggal_string,
    TO_CHAR(tinggi_badan) AS tinggi_char,
    TO_NUMBER('54321') AS contoh_number,
    TO_DATE('2026-10-15', 'YYYY-MM-DD') AS contoh_date
FROM mahasiswa;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 15 in 0.066 seconds

| | NAMA | TANGGAL_STRING | TINGGI_CHAR | CONTOH_NUMBER | CONTOH_DATE |
|---|---|---|---|---|---|
| 1 | Tegar Septian | 28-02-2002 | 174 | 54321 | 15-OCT-26 |
| 2 | Andra Wijaya | 15-02-2004 | 172 | 54321 | 15-OCT-26 |
| 3 | Bunga Lestari | 10-05-2003 | 160 | 54321 | 15-OCT-26 |
| 4 | Candra Pratama | 22-11-2002 | 178 | 54321 | 15-OCT-26 |
| 5 | Dina Amelia | 05-01-2004 | 158 | 54321 | 15-OCT-26 |
| 6 | Eko Saputra | 17-08-2001 | 165 | 54321 | 15-OCT-26 |
| 7 | Fani Rahmawati | 30-12-2003 | 162 | 54321 | 15-OCT-26 |
| 8 | Gilang Ramadhan | 11-04-2002 | 170 | 54321 | 15-OCT-26 |
| 9 | Hana Salsabila | 08-07-2004 | 155 | 54321 | 15-OCT-26 |
| 10 | Irfan Hakim | 25-10-2001 | 175 | 54321 | 15-OCT-26 |
| 11 | Jihan Fahira | 14-03-2003 | 164 | 54321 | 15-OCT-26 |
| 12 | Kevin Julio | 09-09-2002 | 180 | 54321 | 15-OCT-26 |
| 13 | Laila Fitri | 18-06-2004 | 159 | 54321 | 15-OCT-26 |
| 14 | Rizky Febian | 01-12-2001 | 168 | 54321 | 15-OCT-26 |
| 15 | Nadia Vega | 21-08-2003 | 161 | 54321 | 15-OCT-26 |

**d. Fungsi Numerik**

```
SELECT
    nama,
    ABS(-15) AS nilai_abs,
    MOD(berat_badan, 4) AS sisa_bagi,
    FLOOR(tinggi_badan/2.5) AS floor_hasil,
    POWER(3,4) AS pangkat,
    ROUND(berat_badan/4, 2) AS pembulatan,
    CEIL(tinggi_badan/5) AS pembulatan_atas
FROM mahasiswa;
```

Script Output | Query Result

All Rows Fetched: 15 in 0.018 seconds

| | NAMA | NILAI_ABS | SISA_BAGI | FLOOR_HASIL | PANGKAT | PEMBULATAN | PEMBULATAN_ATAS |
|---|---|---|---|---|---|---|---|
| 1 | Tegar Septian | 15 | 1 | 69 | 81 | 17.25 | 35 |
| 2 | Andra Wijaya | 15 | 1 | 68 | 81 | 16.25 | 35 |
| 3 | Bunga Lestari | 15 | 2 | 64 | 81 | 12.5 | 32 |
| 4 | Candra Pratama | 15 | 2 | 71 | 81 | 17.5 | 36 |
| 5 | Dina Amelia | 15 | 0 | 63 | 81 | 12 | 32 |
| 6 | Eko Saputra | 15 | 0 | 66 | 81 | 15 | 33 |
| 7 | Fani Rahmawati | 15 | 3 | 64 | 81 | 13.75 | 33 |
| 8 | Gilang Ramadhan | 15 | 0 | 68 | 81 | 17 | 34 |
| 9 | Hana Salsabila | 15 | 1 | 62 | 81 | 11.25 | 31 |
| 10 | Irfan Hakim | 15 | 0 | 70 | 81 | 18 | 35 |
| 11 | Jihan Fahira | 15 | 2 | 65 | 81 | 13.5 | 33 |
| 12 | Kevin Julio | 15 | 3 | 72 | 81 | 18.75 | 36 |
| 13 | Laila Fitri | 15 | 1 | 63 | 81 | 12.25 | 32 |
| 14 | Rizky Febian | 15 | 2 | 67 | 81 | 15.5 | 34 |
| 15 | Nadia Vega | 15 | 0 | 64 | 81 | 13 | 33 |